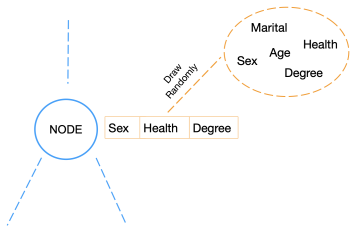
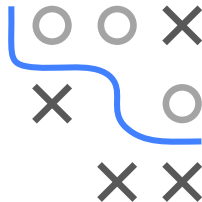


Random Forest In a Nutshell



- Understand basic concept of random forest
- Know basic aggregation rules
- Understand concept of feature importance



LEARNING AND PREDICTION WITH RF I

Prediction

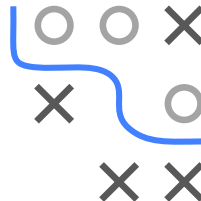
Input: Unlabeled data

Age	Health	Degree of Graduation	Level of Happiness
41	Fair	Bachelor	?
35	Good	Bachelor	?
22	Fair	High School	?

Random
Forest
Model

Prediction

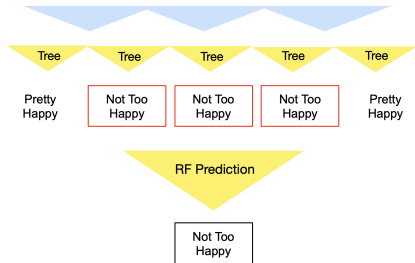
Level of Happiness
Not too happy
Pretty happy
Not too happy



AGGREGATION RULES FOR DIFFERENT TASKS I

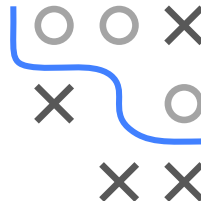
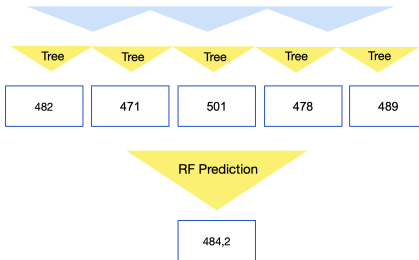
Classification Task - Majority Vote

Age	Health	Degree of Graduation	Level of Happiness
41	Fair	Bachelor	?



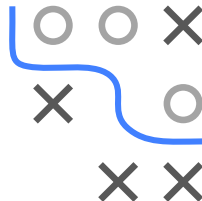
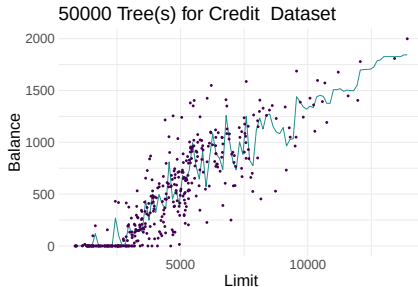
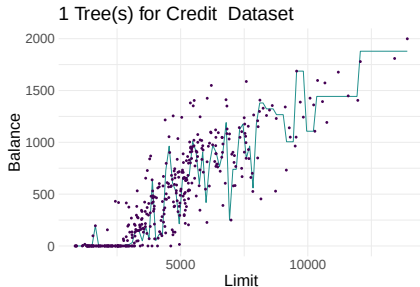
Regression Task - Averaging

Rating	Income	Credit Limit	Credit Card Balance
107	32.318	4351	?



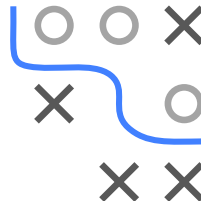
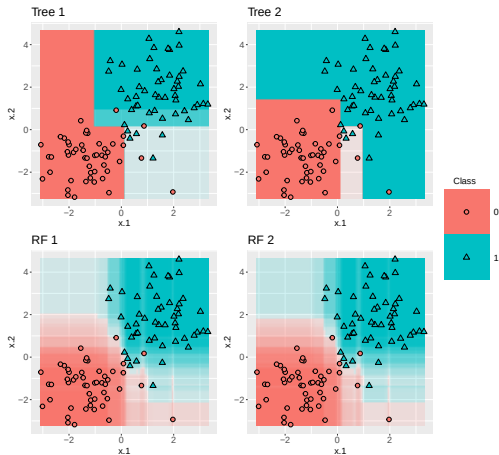
PERFORMANCE OF RF I

- In general: Increasing the ensemble size stabilizes the predictions
 - For regression tasks the stabilization is often not sufficient.



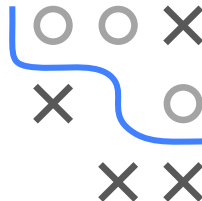
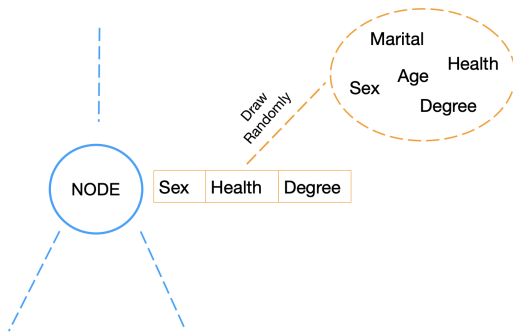
PERFORMANCE OF RF I

- RF performs well for classification tasks:
 - Two different trees → Quite different decision regions
 - Two different RFs → Similar decision regions



PERFORMANCE OF RF I

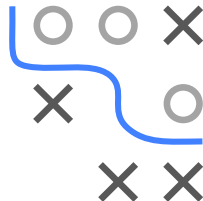
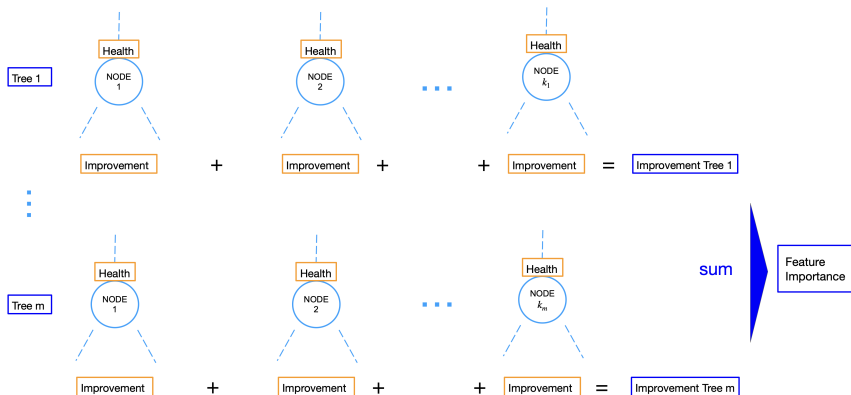
- Trees should be decorrelated, i.e., make mistakes in different directions
- Avoid correlation by
 - Bootstrap sampling
 - Randomized splits. In each node of each tree, consider different features for splitting:



FEATURE IMPORTANCE

Several options, e.g., measure contribution of feature to model:

- Measure based on improvement in splitting criterion
- E.g. Feature importance of 'Health', search all nodes with 'Health' as splitting variable:



A 3x3 grid with a blue path and obstacles. The path starts at the top-left corner (row 1, column 1) and ends at the bottom-right corner (row 3, column 3). The path is composed of three segments: a vertical segment from (1,1) to (2,1), a horizontal segment from (2,1) to (2,2), and a diagonal segment from (2,2) to (3,3). Obstacles are represented by 'X' marks at (1,3), (2,1), (3,1), and (3,2). There are also empty cells at (1,2), (2,3), and (3,3).

-
- The diagram illustrates the process of computing OOB Loss for each observation using permutation. It shows two data tables for 'Age', 'Health', 'Degree of Graduation', and 'Level of Happiness'. These are permuted to create 'Tree 1' through 'Tree m'. For each tree, the OOB loss is calculated for each observation. The final result is the 'OOB Permutation Loss of Observation i', which is the average of the OOB losses across all trees.
- Original Data Table 1:**
- | Age | Health | Degree of Graduation | Level of Happiness |
|-----|-----------|----------------------|--------------------|
| 23 | Fair | Bachelor | Not too happy |
| 23 | Fair | Bachelor | Not too happy |
| 48 | Good | High School | Very happy |
| 70 | Excellent | High School | Pretty happy |
- Permuted Data Table 1 (Tree 1):**
- | Age | Health | Degree of Graduation | Level of Happiness |
|-----|-----------|----------------------|--------------------|
| 23 | Good | Bachelor | Not too happy |
| 23 | Excellent | Bachelor | Not too happy |
| 48 | Fair | High School | Very happy |
| 70 | Fair | High School | Pretty happy |
- Permuted Data Table 2 (Tree m):**
- | Age | Health | Degree of Graduation | Level of Happiness |
|-----|-----------|----------------------|--------------------|
| 70 | Excellent | High School | Pretty happy |
| 48 | Good | High School | Very happy |
| 70 | Excellent | High School | Pretty happy |
| 23 | Fair | Bachelor | Not too happy |
- Observation i:**
- | Age | Health | Degree of Graduation | Level of Happiness |
|-----|--------|----------------------|--------------------|
| 23 | Fair | Bachelor | Not too happy |
- OOB Loss Calculation for Observation i:**
- For Tree j, the OOB loss is calculated as:
- | Prediction | True Label | Loss |
|------------|---------------|------|
| Very happy | Not too happy | 0.89 |
- For Tree r, the OOB loss is calculated as:
- | Prediction | True Label | Loss |
|--------------|---------------|------|
| Pretty happy | Not too happy | 0.58 |
- OOB Permutation Loss of Observation i:**
- The final OOB loss for Observation i is the average of the OOB losses across all trees (Tree 1 to Tree m).

FEATURE IMPORTANCE I

